

End-to-End AI Video Studio Architecture

Shine — Next-Gen AI Micro-Drama Studio

Transforming raw text prompts into viral 9:16 vertical drama series (20–50 episodes) with multi-agent AI directors, consistent character personas, in-browser WebGL editing, and Google Cloud serverless infrastructure.

MARKET SEGMENT

Vertical Micro-Drama

\$10B+ Global Market Boom

CREATIVE BRAIN

Multi-Agent AI Director

Gemini 3.5 Flash & Veo 3.1

Editor Core

Pixi.js WebGL NLE

In-Browser Zero-Cost Render

INFRASTRUCTURE

Cloud Run Serverless

Scale-to-Zero (\$0 Idle Cost)

MARKET CONTEXT

The Explosive Rise of Vertical Micro-Dramas

\$10B+

GLOBAL MARKET BY 2027

Pioneered by Chinese Wēi Duǎnjù and rapidly expanding across the US, Southeast Asia, Latin America, and Europe via TikTok, ReelShort, DramaBox, and Shorts.

1–3 Min

EPISODIC FORMULA

Serialized 20–50 episode dramas with fast conflict escalation, intense emotional payoffs, and cliffhangers every 60–90 seconds to maximize viewer retention.

3.5x

HIGHER MONETIZATION VELOCITY

Micro-dramas convert viewers into paying subscribers 3.5x faster than traditional streaming platforms through pay-per-episode paywalls.

 **Opportunity:** The creator economy needs automated software to produce hundreds of high-quality vertical drama episodes per month with minimal budget.

MARKET PAIN POINTS

Core Production Bottlenecks in Vertical Drama

1. High Physical Production Costs

Traditional filming requires casting actors, renting sets, filming crews, and extensive editing suites, costing \$30,000–\$100,000+ per series.

2. Character Visual Drift in Generative AI

Existing text-to-video tools fail to preserve consistent facial geometry, attire, and lighting across consecutive scenes, ruining narrative immersion.

3. Prohibitive Localization & Dubbing Costs

Translating video series into foreign languages traditionally requires re-rendering video clips from scratch, incurring massive GPU compute bills.

4. Disconnected Toolchains

Creators waste hours juggling separate tools for scriptwriting (ChatGPT), image gen (Midjourney), video gen (Runway), and NLE editors (Premiere).

TARGET MARKET

Product-Market Fit & Ideal Customer Profiles

Independent Writers & Creators

Web novel authors and solo creators who want to transform written stories into monetizable vertical video series without camera crews.

Value: 100x faster production

Digital Media Studios & Agencies

Agencies producing high-volume serialized content for TikTok, YouTube Shorts, and Reels to capture ad revenue and subscriber paywalls.

Value: 80% cost reduction

Global Localization Teams

Publishers localizing domestic drama hits into international markets (EN, VI, ZH, JA, KO, ES) with instant neural voice swapping.

Value: Zero-video-render dubbing

 **The Shine Value Proposition:** Turn a single creative individual or small team into a full-fledged serialized drama production studio.

SYSTEM BLUEPRINT

High-Level System Architecture & 5-Tier Pipeline

1. Client WebGL Studio

ACTIVE

Vue 3 SPA + Pixi.js v8 WebGL Canvas NLE. Zero-render real-time scrub & in-browser WebCodecs MP4 export.

2. Cloud API Gateway

ACTIVE

Node 22 Express :3001 + JWT Auth & 31 REST Controllers orchestrating multi-agent pipelines.

3. AI Director Brain

ACTIVE

Gemini 3.5 Flash & Veo 3.1 + Parallel AI Search MCP & Grafana MCP for viral trends & telemetry.

4. Serverless Workers Link

ACTIVE

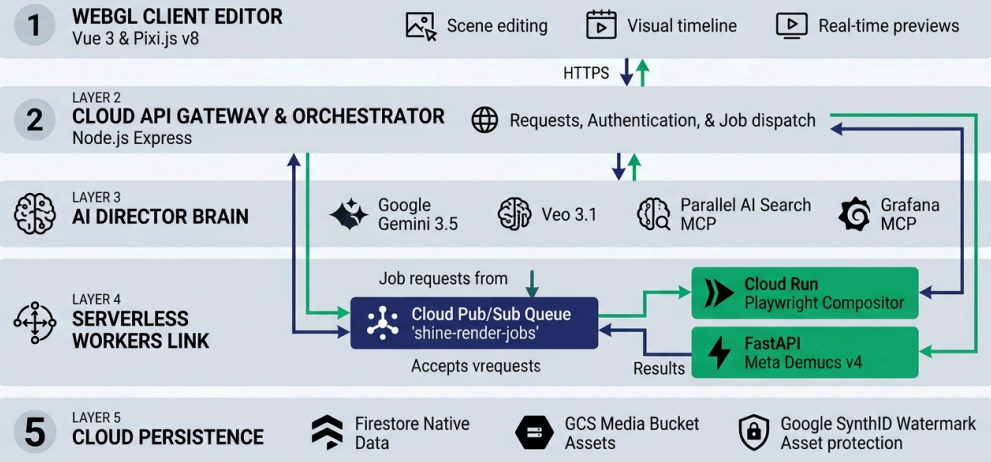
Cloud Pub/Sub (shine-render-jobs) dispatches tasks to Cloud Run Playwright Compositor & FastAPI Demucs v4 AI.

5. Data & Storage

ACTIVE

Firestore Native (shine-db), GCS Signed URLs & Google SynthID steganographic watermarking.

'Shine AI Micro-Drama Studio'



WORKER BUS: Cloud Pub/Sub (shine-render-jobs) • REST/SSE • MCP Protocols **100% Google Cloud Native**

END-TO-END EXECUTION FLOW

Agentic Micro-Drama Generation Pipeline Workflow

1. Viral Trend DiscoveryAI SEARCH

TrendRadar scans viral drama tropes across 6 global regions via Parallel AI Search MCP.

2. Master PlanGEMINI 3.5

StorySkeleton designs 20–50 episode narrative master plans with cliffhangers per episode.

3. Compliance & SafetyAUDIT

SupervisionAgent verifies copyright safety, content policy, and pacing quality gates.

4. Episode BreakdownSCRIPT

ScriptAgent outputs 15–45 scene blocks (4–8s each) with visual prompts & dialogue in JSON.

5. Resource Gen B1–B6PARALLEL

B1 Persona · B2 Veo 3.1 Video · B3 TTS · B4 Demucs v4 · B5 Lyria 3 BGM · B6 SynthID

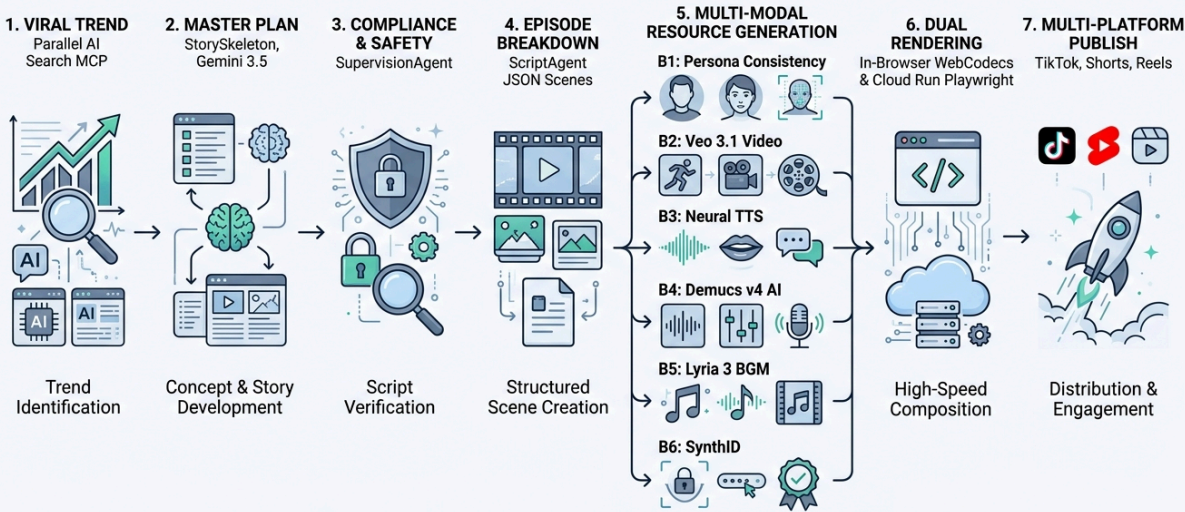
6. Dual RenderingRENDER

In-Browser WebCodecs (0\$) or Cloud Run Playwright Compositor for high-quality export.

7. Multi-Platform PublishPUBLISH

Distribute completed 9:16 MP4 episodes to TikTok, YouTube Shorts, and Instagram Reels.

A Complete End-to-End Agentic Micro-Drama Generation Pipeline Workflow



LAYER 1: PRESENTATION (ACTIVE CORE)

OpenVideo Pixi.js WebGL & WebCodecs Engine

1. Real-Time WebGL Compositor (Pixi.js v8)

The browser viewport renders 9:16 vertical frames at ≥ 30 FPS using client GPU hardware.

- **Zero-Render Preview:** Scrub playhead across video, audio, and subtitle buffers with 0ms cloud latency.
- **GLSL Fragment Shaders:** Real-time chroma keying, color grading, and dynamic transition blends.
- **Microsecond Timeline Model:** Sub-millisecond timing precision (1s = 1,000,000 μ s).

2. In-Browser Export (WebCodecs API)

Renders MP4 video directly on the user's computer via hardware-accelerated WebCodecs ('mediabunny').

- **Zero Server Cost (\$0):** 100% computed on client device, saving thousands in GPU hosting bills.
- **Lightning Speed:** Encodes a 90-second 1080p episode in about 12 seconds.
- **In-Memory ISOBMFF Muxing:** Assembles the final MP4 container in memory for immediate download.

PLANNED ROADMAP SPECIFICATION (PHASE 2)

Real-Time Team Collaboration Architecture

Delta Patch Protocol (`PatchSyncService`)

Designed to broadcast atomic RFC 6902 delta patches (< 1KB) over Socket.io instead of heavy 500KB blobs:

```
{ "op": "update", "path": "/clips/clip_01/timing/display/to", "value":  
4500000, "author": "usr_sarah" }
```

Planned for co-editing sessions with sub-50ms peer update targets.

Planned Concurrency Controls

- **Clip-Level Mutex Locking:** Optimistic locking to prevent race conditions when multiple users edit the same clip.
- **Live Teammate Cursors:** Playhead markers broadcasting peer listening positions at 30Hz.
- **Immutable Version Checkpoints:** Continuous timeline revision snapshots saved to Firestore.

Status: Architecture & Backend Service Designed → Full Client Integration in Phase 2.

LAYER 3: AI BRAIN (ACTIVE CORE)

Hierarchical Multi-Agent AI Director

1. Discovery & Skeleton

TrendRadar: Scans viral drama trends via Parallel MCP.

StorySkeleton: Designs 20–50 episode narrative master plans with cliffhangers.

Model: gemini-3.5-flash-lite

2. Screenplay Breakdown

ScriptAgent: Breaks each episode into 15–45 short scene blocks (4–8s) with visual prompts, actions, and dialogues in JSON.

Model: gemini-3.5-flash-lite

3. QA & In-Editor Copilot

SupervisionAgent: Audits pacing and safety.

ChatbotAgent: Modifies timeline state via natural language commands.

Model: gemini-3.5-flash-lite / 2.5-pro

 **4-Tier Memory Mesh:** Active Session State → Vertex Vector Search RAG (`text-embedding-004`) → Series Knowledge Graph Lineage → Token Compressor (compresses 500KB JSON into ~4KB Markdown prompts).

LAYER 3: LOCALIZATION (ACTIVE CORE)

Decoupled Audio-Visual Dubbing Engine

Track 1: Pure Visual Motion (VIDEO 1)

Generated as silent video clips via Google Veo 3.1 (`veo-3.1-generate-001`).
Immutable during multi-language dubbing operations.

[Scene 01: 6.00s] — [Scene 02: 5.50s] (Fixed Visuals)

Track 2: Neural TTS Dialogue (AUDIO 1)

Synthesized per language track via `gemini-3.1-flash-tts-preview` (30 distinct voices) with word-level phonetic timestamps.

EN: "Where are you?" (2.1s) → VI: "Anh đang ở đâu?" (2.6s, Δt)

💡 **Unfair Cost Advantage:** To distribute in 5 languages, Shine only swaps the voice track and recalculates timeline timings (Δt). **You never re-render expensive video AI clips!**

LAYER 4: CLOUD WORKERS (ACTIVE CORE)

Serverless Cloud Run Video Render Worker

1. Asynchronous Jobs

Episode render tasks are dispatched asynchronously (`/render-job`), tracking progress from 0% to 100% via API polling or SSE stream.

2. Headless Playwright

`shine-render-worker` container launches headless Chromium WebCodecs compositors to render master-quality 1080p/4K video frames.

3. Scale-to-Zero (\$0 Idle)

Cloud Run containers scale automatically to 0 instances when idle, eliminating static server compute expenses.

 **Parity Audit:** Automated SSIM diff tool ensures the final cloud video matches the editor preview with **SSIM > 0.999 parity**.

LAYER 4: AUDIO AI (ACTIVE CORE)

Meta Demucs v4 AI Stem Separation & 3D Audio

FastAPI Demucs v4 AI Microservice

A dedicated FastAPI Python microservice on Cloud Run (``services/demucs-worker``) separates mixed audio into isolated stems:

- **Vocal Isolation:** Separates character dialogue from background ambient noise.
- **Music Extraction:** Strips vocal tracks to create clean instrumental soundtracks.
- **Parametric Auto-Ducking:** Attenuates BGM by 80% with 250ms attack during dialogue intervals.

3D Binaural Spatial Audio Engine

Calculates real acoustic physics based on where characters stand on the phone screen:

- **Sabine RT60 Decay:** Simulates realistic room reverb (hallway, bedroom, street).
- **Woodworth Head Model:** Computes Interaural Time Differences (ITD) between ears.

Cinema-quality surround sound directly through standard headphones.

LAYER 5: STORAGE & TRUST (ACTIVE CORE)

Pluggable Persistence, SynthID & C2PA Provenance

Pluggable Database Architecture (IDatabaseProvider)

Enforces strict repository interfaces, enabling database switching via environment variables:

Firestore Native (`shine-db`) – Primary Cloud NoSQL

MongoDB Atlas – Document Database

SQLite (`shine.db`) – Local Development

Digital Watermarking & Provenance

- **Google SynthID:** Embeds imperceptible digital watermarks into synthesized audio waveforms and video frames.
- **C2PA Content Credentials:** Cryptographically signs provenance manifests declaring AI generative origin.
- **V4 Signed URLs:** Generates temporary 30-minute GCS links for secure media streaming.

INFRASTRUCTURE SYNERGY

Google Cloud Unified Service Synergy

Compute & Scaling

Cloud Run: Serverless execution with `--min-instances 0`` scaling to 0 when idle.

Cloud Scheduler: Periodic cron (``*/5 * * * *`) token sync heartbeat.

\$0 Idle Compute Cost

Data & Messaging

Firestore Native: Document persistence for scripts, timelines & version snapshots.

Cloud Storage (GCS): V4 signed URLs for high-throughput media streaming.

gs://shine-studio-media

Security & Observability

Cloud IAM & Secret Manager: Least-privilege roles (``aiplatform.user``, ``datastore.user``).

Cloud Logging: Real-time structured telemetry and P99 latency tracking.

Enterprise IAM & Audit Trails

 **Synergy Advantage:** Native identity & networking across all GCP services eliminates ingress/egress transit bottlenecks and simplifies DevOps automation.

GENERATIVE AI STACK

Multimodal Generative AI Model Matrix

Domain / Task	Active Production Model	Supported Models & Fallbacks	Technical Role in Shine
Script & Planning	`gemini-3.5-flash-lite`	`gemini-2.5-pro`, `gemini-3.1-pro`	Story master plan, 15-45 scene JSON breakdown, cliffhangers.
Agent Orchestration	`gemini-3.5-flash-lite`	Multi-Agent ADK Router	TrendRadar, ScriptAgent, SupervisionAgent pipeline coordinator.
Video Generation	`veo-3.1-generate-001`	`veo-3.0`, `veo-2.0`, `veo-2.1`	9:16 vertical silent video motion clips with camera trajectory steering.
Facial Consistency	`gemini-3.1-flash-lite-image`	`imagen-3.0-generate-002`, `imagen-3.5`	8-angle facial consistency keyframes and character visual DNA.
Neural TTS Speech	`gemini-3.1-flash-tts-preview`	30 Gemini Voices (`Zephyr`, `Puck`...)	Multi-speaker dialogue with word-level phonetic timestamps in 6 languages.
Live Audio Dialogue	`gemini-live-2.5-flash-native-audio`	Bidirectional WebSocket Stream	Low-latency live interactive character voice conversations & real-time dubbing.
Soundtrack & SFX	`lyria-3-clip-preview`	`Lyria-v1`	Dynamic mood-adaptive background music & suspense crescendo risers.
Vector Search (RAG)	`text-embedding-004`	Vertex Vector Search	Sub-50ms semantic search across scene bibles and character lineage.
Stem Separation	`Meta Demucs v4 (htdemucs)`	FastAPI PyTorch Container	Vocal isolation, BGM extraction, and parametric auto-ducking.
Watermarking	`Google SynthID`	C2PA Manifest Signing	Steganographic imperceptible watermarking on generated waveforms.

PRIMARY API: Official `@google/genai` Node.js SDK

FALLBACK: Google Flow Session Account Pool (flowST)

PROTOCOL INTEGRATION

Model Context Protocol (MCP) Integration Fabric

1. Parallel AI Search MCP (`search.parallel.ai/mcp`)

ACTIVE

Connects Shine's multi-agent director to real-time web search for discovering viral drama tropes and performing safety audits:

- **Trend Radar Agent:** Scans real-time trending micro-drama tropes across 6 regions.
- **Supervision Agent:** Automated pre-flight copyright check on titles, character names, and novel synopses.
- **SfxService:** Discovers open sound effects across Freesound, Pixabay, and web sources.

```
Client: `server/src/integrations/mcp/ParallelMCPClient.ts`
```

2. Grafana MCP & Loki Observability

ACTIVE

Unifies live system monitoring, container health, and error tracing through the official Grafana MCP endpoint:

- **Telemetry Streaming:** Auto-flushes structured logs and memory RSS (MB) to Grafana Cloud Loki every 15s.
- **P99 Latency & Probes:** Exposes `GET /api/admin/observability` tracking active WebSocket connections.
- **Automated Alerts:** Triggers instant email alerts upon critical AI pipeline failures.

```
Service: `server/src/services/observability/GrafanaObservabilityService.ts`
```

🔧 **Standardized Tool Calling:** Open MCP protocol ensures modular tool execution across Google Agent ADK without vendor lock-in.

COMPETITIVE EDGE

Shine's 4 Unique Selling Points (USPs)

1. Decoupled Multi-Language Dubbing

Silent visual clips + neural speech allow swapping dialogue into 6+ languages and auto-aligning timeline bounds (Δt) without re-rendering expensive video AI clips.

2. In-Browser \$0 GPU WebCodecs Render

Creators export 1080p MP4 episodes directly in browser memory in ~12 seconds, delivering fast turnarounds with zero server GPU overhead.

3. Persona Studio & 8 Facial Anchors

Locks facial geometry, wardrobe, and character embeddings across 50 episodes, eliminating visual drift and inconsistency.

4. Scale-to-Zero Serverless Economy

100% serverless infrastructure on Google Cloud Run automatically scales containers to 0 instances when idle, ensuring \$0 static maintenance costs.

CONSISTENCY ENGINE

Persona Studio: 8 Facial Consistency Anchors

8-Angle Facial Consistency Locking

When a character is registered, Persona Studio generates 8 multi-angle reference keyframes to lock their biometric DNA:

- | | |
|---------------------|-----------------------|
| • Frontal Portrait | • 45° Left Profile |
| • 45° Right Profile | • Full Side Profile |
| • High Angle Shot | • Low Angle Cinematic |
| • Emotional Smirk | • Dramatic Anger |

Prompt DNA & Reference Injection

During video and storyboard generation, the character's visual traits and reference image URLs are automatically injected into Gemini and Veo prompts.

- Preserves signature hairstyles, scars, and attire continuity.
- Links voice IDs (Gemini Audio) to specific character personas.

Guarantees cast continuity across all 20-50 episodes.

AUDIENCE RETENTION

Kinetic Subtitles & Dynamic Cliffhanger Engine

Word-Level Animated Kinetic Captions

Captions drive over 60% of short-form video watch time. Shine automatically generates viral animated subtitles:

- **Karaoke Pop-In:** Active words highlight and bounce on beat with speech.
- **Typography Presets:** Dramatic Punch, Neon Cyberpunk, Clean Minimalist.
- **One-Click Translation:** Translates subtitles into foreign languages while preserving millisecond timings.

Dynamic Cliffhanger Engine

Automatically identifies the episode climax and injects an intense 3-second hook sequence:

- **GLSL Shaders:** Triggers glitch, flash, or keyframe zoom-in effects.
- **Audio Stinger:** Injects dramatic crescendo risers on `AUDIO 3`.
- **Call-to-Action Overlay:** Displays high-impact prompt: *"Episode 2 Unlocked in 3s"*.

Maximizes episode completion rates and subscription paywall conversion.

DEVELOPMENT

EVOLUTION

Agile Engineering Roadmap (Sprints 1-7)

Sprint 1: Foundation, Vertex AI SDK, Pluggable DBs (Firestore/SQLite), and Auth Suite.	v0.1 Alpha
Sprint 2: Multi-Agent Script Pipeline, Trend Radar (Parallel MCP), Persona Studio.	v0.2 Beta
Sprint 3: OpenVideo Pixi.js WebGL Editor, Undo/Redo Commands, Dual-Render Parity.	v0.5 NLE Beta
Sprint 4: Neural Voiceover, Kinetic Subtitles, Cliffhanger Engine, Meta Demucs v4.	v0.8 RC1
Sprint 5: WebSocket Delta Sync Architecture, Quality Linter & i18n Key Parity.	v0.9 RC2
Sprint 6 & 7: Multi-Platform Publishing, Grafana MCP Observability & Playwright E2E Suite.	v1.1 Commercial

FUTURE PLANNING (PHASE 2 & 3 ROADMAP)

Planned Roadmap Innovations

1. Real-Time Multi-User Co-Editing

Full client integration of the Socket.io RFC 6902 JSON Pointer delta patch engine for live collaborative timeline editing.

Phase 2: Live Team Collaboration

2. Distributed Batch Render Queue

Automated multi-episode batch queueing across a pool of parallel Cloud Run Playwright worker nodes via Cloud Pub/Sub.

Phase 2: Automated 50-Episode Batch Queue

3. Platform Analytics & Synthetic Audience

Viewer retention heatmaps, drop-off prediction curves, and cross-platform publishing analytics (TikTok / Shorts / Reels).

Phase 3: Platform Analytics & Audience Simulator

4. Live Interactive Drama & Marketplace

Branching narrative trees ('@antv/g6') for live voting + AI character asset trading marketplace.

Phase 3: Branching Drama & Character Hub

COST ARCHITECTURE

Cloud Run Sizing & \$0 Idle Cost Model

Service	Specs	Autoscaling	Concurrency	Idle Cost
`shine-app`	2 vCPU / 4Gi	0 to 3 instances	80 req/instance	\$0.00
`shine-render-worker`	4 vCPU / 8Gi	0 to 3 instances	1 job (Isolated)	\$0.00
`demucs-worker`	2 vCPU / 4Gi	0 to 3 instances	2 req/instance	\$0.00

 **Scale-to-Zero Guarantee:** When no users are actively editing or rendering, all 3 Cloud Run services scale to 0 instances. Zero monthly server bills when idle!

DEVOPS & AUTOMATION

1-Click Full Ecosystem Cloud Run Deployment

Automated Cloud Provisioning Lifecycle

1. Auto-enables 10 GCP APIs (`run`, `pubsub`, `firestore`, `aiplatform`, etc.).
2. Grants least-privilege IAM roles to Default Compute Service Account.
3. Auto-provisions Firestore Native database `shine-db` and GCS bucket.
4. Creates Pub/Sub topics (`shine-render-jobs`, `shine-render-status`).
5. Builds and deploys all 3 containers with dynamic YAML configuration.

Single Execution Command

```
# Windows PowerShell .\scripts\deploy-cloudrun.ps1 # Linux / macOS /  
Cloud Shell ./scripts/deploy-cloudrun.sh
```

```
Health Check: `curl https:///api/health`
```

Strategic Summary

Redefining Global Storytelling for the Short-Form Video Era

Shine unites cutting-edge Google generative AI models (`gemini-3.5-flash-lite`, `veo-3.1`, `gemini-live-2.5`, `lyria-3`), Model Context Protocol integrations (Parallel MCP & Grafana MCP), and serverless scale-to-zero infrastructure to turn any creator into a global micro-drama studio.

100x Faster

Script to Video

80% Cheaper

Zero-Render Dubbing

\$0 Idle Cost

Serverless Run